

ELSEWEDY GENZ CODERS ACADEMY

ROBOT BUILD

Hardware & Arduino Mastery

AGE GROUP

Ages 13+

TOTAL SESSIONS

12

REGISTRATION

600 EGP

Your Vision: Bridge the gap between software and physics. Build, wire, and program a live robot.

COURSE CAPSTONE PROJECTS

- Autonomous Robot: Implement movement logic and motor drivers.
- Interactive Circuit Wiring: Master breadboarding and circuit series/parallel logic.
- Robot Demo Showcase: Present your bot's autonomous navigation.

12-SESSION STRUCTURE (ROADMAP)

#	Session Title (Engineer-Led)	Project/Outcome (CTA Supported)
1	Intro to Electronics	Voltage, current, resistance (Core concepts)
2	Components + Circuits	Breadboard: LEDs, resistors (Series/Parallel)
TS 1	Circuit Building Lab	Free build on breadboard (Confidence)
3	Sensors + Microcontrollers	Buttons, sensors + Arduino intro (Digital input)
4	Programming Outputs	Blink LED, control motor (Code to hardware)
TS 2	Sensor + Code Exercises	Read inputs, control outputs (Applied coding)
5	Motor Control	Motor drivers + movement logic (Actuation)
6	Robot Design + Planning	Plan your robot project (Engineering thinking)
TS 3	Design Review	Finalize plan (Readiness)
7	Build Your Robot	Assemble + wire components (Integration)
8	Test, Debug & Present	Fix, improve, demo robot (Real product)
TS 4	Robot Demo Day	Present to Group & Showcase

All 12 sessions alternate between Engineer-Led Core instruction and CTA-Led Technical Support for 1-on-1 guidance.

Partnership Pricing: Ask your school coordinator! Students from partner schools pay only **510 EGP** (15% off standard registration).

CODE IT. BUILD IT. OWN IT.

Launching 8 June 2026 | @elsewedy.genz.coders | www.genzcoders.academy